

Structural-residual reduction chain: the whole "quantum gravity" question collapses to one falsifiable physical statement -- is the seam (E8)_1 at $\tau=i$?

start

Naïve residual

"build a quantum-gravity measure"

v175

Net existence + full-cone RP

discharged to [E] (CAR functor)

v176-v181

One geometric premise

QGEO.SYM.01: carrier μ_4 clock = seam conformal deck

v194-v201

Non-circular form

state-invariance $w \circ \rho = w$; DtN mark-local (Z4)

v234-v235

ONE condition: holomorphy

no abelian sector $\Leftrightarrow \det K=1$ (the Kitaev E8 tower)

v276

Flat all-orders closure (Lean)

flat $\tau=i \Rightarrow [\rho, H]=0$ to ALL orders (FORM.QGEO.03)

v282

Two faces, ONE object

$\chi_{E8(i)}=12$: flat $\tau=i$ geometry = (E8)_1 holomorphy

v284-v285

Two routes, one open lemma

RP-uniqueness $5/6$ + condensation $3/4$; open lemmas coincide

v286-v288

SEAM.EQUIV.01 named + attacked

import firewall (v286); Route A $4/5$ (v287); Route B proves full-L2 Z4 lift (v288)

v289-v297

Flat-Away: 3 routes reduced

heat a_2 closed+Lean (v292/v295/v296), spectral Hessian PD (v293), Troyanov (v294); Route A citable stack (v297)

v300-v302

Closing arc: shared fact pinned

Flat-Away hard+pin from (E8)_1 Steklov (v300); Route A invertible via free fermions (v301); gap = derived $6\ln(3/2)>0$ (v302)

v335-v379

Closed modulo cited theorems

QGEO.SYM.01 = corollary of SEAM.EQUIV.01 (v335); gapped lattice model (v367/v368) + S3 stack (v376-v379), Lean-pinned MMST

v458-v480

Certification round

extension leg = crossed-product package, realisation at invariant level (v469); R3 bridge = Kronheimer 1989 (v479); four-interval μ_4 mechanism (v480)

BEDROCK

No TFPT-internal assumption left

SEAM.EQUIV.01 residual = cited continuum scaling-limit existence (v336) over established facts; stays [0] (not machine-proved end-to-end)

Everything above is a theorem, a Lean proof or an established citation; the bedrock, once a definition (QGEO.SYM.01), is now one falsifiable physical question: is the raw seam (E8)_1 at $\tau=i$?